Philosophical Studies
https://doi.org/10.1007/s11098-025-02430-9

The antinomy of artificial art

Joseph G. Moore[1]

Received: 21 August 2025 / Accepted: 5 October 2025
© The Author(s), under exclusive licence to Springer Nature B.V. 2025

Abstract
This paper explores the tensions that arise in applying our received notions of artistic creation and credit to AI-generated art. Focusing on AI-generated images, I articulate an "antinomy of artificial art"—a contradiction between three plausible propositions: (1) AI-aided images can count as works of art, (2) All artworks have artists, and (3) AI-aided images don't (clearly) have artists. To support (3), I argue that, in the case of many AI-aided works, none of the potential candidates—the AI system itself, its programmers, the creators of training images, or the human prompter—can be unproblematically credited as artist. I explore three intriguing responses to this antinomy: treating the prompter as an institutional artist, crediting the AI as a creative agent, and viewing AI–human collaboration as an artistic collective. While each approach shows promise, I argue that none succeeds. If I'm right, current generative AIs inhabit a sort of conceptual limbo. They are "super-tools" possessing sophisticated generative capacities but lacking the robust intentionality and normative responsiveness required for prototypical artistic creation. The tension they expose in our conceptual scheme can't be resolved through further empirical or theoretical investigation, I suggest. Rather it will be resolved through a socially-driven revision of our artistic concepts and institutions as we adjust them to our developing AI-aided artistic practices.

Keywords Artificial intelligence · AI-generated art · Art · Creativity · Authorship · Institutional theory of art · Generative AI · Machine learning · Agency · Artistic collectives · Copyright and intellectual property · Aesthetics · Intentionality · Normativity

✉ Joseph G. Moore
jgmoore@amherst.edu

[1] Philosophy Department, Amherst College, P.O. Box 5000, Amherst, MA 01002, USA

Published online: 23 October 2025

1 Introduction

Artificial intelligence is transforming the ways we make—and think about—art. Generative AIs produce aesthetically compelling and seemingly creative works in a number of different artistic domains. And they can do so almost on their own. This challenges the ways we think about creativity and artistic credit by pushing up against a "one-artwork–one-artist paradigm" that lies at the core of our conceptual scheme: in the prototypical case, a single human artist engages in an intentionally directed and normatively responsive creative process that results in an artwork. (I touch later on how this paradigm informs even the way we approach collective art-making, such as in film and group performance.) In their sophisticated generative capacities, current AIs seem more than just artistic tools, but (as of 2025) they don't have all the agential characteristics of the proto-typical artist. They lack what I'll call robust intentionality and normative responsiveness. They are super-tools, yet they remain partial agents at best. As such, they inhabit a conceptual limbo that challenges us not just philosophically but legally: recent litigation against AI companies by artists, as well as by major publishers, underscores the practical urgency of this challenge. I won't dwell on the legal issues here.[1] Rather, I'll articulate and develop what I regard as an underlying puzzle art-making AIs pose to our artistic practices and to the conceptual scheme that surrounds them.

Focusing on the visual domain, I present (in Sects. 2 and 3) a representatively ordinary example of an AI-aided image—"Hippie Philosophers*"—that I then use (in Sect. 4) to articulate the puzzle. I motivate a "no artist argument" to the effect that, in at least my ordinary case, there's no clear artist to credit—not the AI developer, not the artists behind ingested images, not the prompter, and not the AI itself. This argument leads to a conceptual tension so puzzling that I grandly articulate as an "antinomy" of artificial art: ordinary AI-aided images can count as works of art; artworks have artists; ordinary AI-aided images don't have artists.

In the remainder of the paper, I develop a dialectic around this antinomy. I consider three nuanced ways one might push back on the antinomy. In Sect. 5, I explore how one might extend an institutional theory of art as a way of crediting a casual prompter as artist. In Sect. 6, I consider whether we can count an AI itself as artist by stretching our concept of artistic creation. And in Sect. 7, I consider the possibility of side-stepping the no-artist argument by holding that AI-aided art is produced by an artistic collective that includes both prompter and AI. Although all three "push-backs" show promise, I argue that none can succeed at resolving the antinomy as it concerns our current concepts and practices. In the face of rapidly improving AI art, perhaps we should, and perhaps we will, revise these concepts and practices. What shape this revision takes is an open question. (The push-backs I discuss point along promising but divergent revisionary paths.) But I suggest, in the final Sect. 8, that any revision will be a matter not of discovery, but of social choice.

[1] For a discussion of the legal issues here, see Moore & Frankel 2024.

2 Inputs and outputs

Although I think my discussion generalizes to all contemporary AI-aided art, and even beyond, to AI-generated content generally, I'll focus here on AIs in the visual domain. In my central example, I invoke a fictional AI, "Cube-E", based upon the workings and capacities of current image-generating AIs such as Dall-E, Midjourney and Stable Diffusion. These AIs ingest close to a billion images during a training process, copying existing works by human artists. The resulting machine learning capabilities are then drawn upon in generating a novel image from a user's prompt. In discussing the conceptual tensions these AIs raise, it will be useful to distinguish between problems of input and problems of output. Although my focus here isn't legal, this distinction is usefully approached through recent battles over copyright.

On the input side, artists and other right holders, like Getty images, have sued AIs for infringement on their copyright to ingested images. (The New York Times and other prominent publishers are now suing other content-producing AIs for similar reasons.) So far, these suits have been largely unsuccessful. The primary legal argument against input-based copyright claims is grounded in transformative fair use: the AIs don't copy ingested artworks in order to use them as-is, rather they copy them as part of a deep learning process yielding a capacity that is in turn engaged in producing novel outputs.[2] Below I'll consider a connected, theoretical argument—namely, that the scope and character of artistic influence upon these AIs is, for the most part, not relevantly different than acceptable artistic influences in traditional art-making. That is, art-making AIs don't borrow any more than human artists have always done.[3]

On the output side, courts have been wrestling with who, or what can claim copyright to AI-aided creations. So far, both humans and the AIs themselves have been shut out. As a prominent example on the human front, Christina Kashtanova created a graphic novel using Midjourney to generate images, initially receiving copyright protection for the entire work. However, the US Copyright Office revoked copyright to the images (but not the text) when it discovered that she had used AI to generate them. And on the AI front, engineer, Stephen Thaler has been unsuccessful so far in having copyright assigned directly to his AI, "Creativity Machine" for one of the images it produced. But these cases are clearly not the end of the turmoil. AIs have pushed us into a legal wild west which reflects, I think, the underlying conceptual tension I'll explore here. With the distinction between problems of input and problems of output in place, I'll now present an extended example that will be the focus of discussion throughout the paper.

[2] This defense follows established precedents in copyright law, such as the Two Live Crew case in which the US Supreme Court ruled that a parody of Roy Orbison's "Pretty Woman" constituted transformative fair use rather than illegal copying.

[3] Both of these arguments are presented in Moore & Frankel 2024. The central example of the next section is also borrowed from this article, as is my presentation of the "old wine" argument.

3 Arta and Theo

Consider an aspiring painter, Arta, who learns technique and develops her stylistic voice by spending days on end in the great museums of Paris. Some days Arta simply looks at paintings, taking in the workings of the great masters. But some days she takes along an easel, as museums allow, copying great works in her own brush as a way of solidifying technique and expression. Arta also spends time studying poster-prints from 1967's Summer of Love. After some years, Arta has found her voice—"psychedelic neo-cubism". One day she paints a colorful painting of two hippie philosophers in discussion, which she titles "*Hippie Philosophers*".

According to our existing legal and artistic concepts, Arta has full copyright to her image. She deserves compensation from anyone who seeks to reproduce her image. And according to our underling intuitions, Arta should get full credit—aesthetic praise or blame—for creating *Hippie Philosophers.* Neither Picasso, nor Braque, nor any of the influential 60's print makers, nor their estates, nor any of the other artists (nor their estates) that influenced Arta's style have any legal or intuitive claim on Arta's creation. Any good biography of Arta ought to mention these formative influences, but *Hippie Philosophers* is unproblematically Arta's, and Arta's alone.

Contrast this with a procrastinating theorist, Theo, who one day idly types "cubist painting of hippie philosophers in discussion" into a (fictional) generative art AI, which I'll call Cube-E. Theo is surprised by how much he likes one of the images that Cube-E spits out, and so he posts it on his website, giving it the label "*Hippie Philosophers**". (It's part of Theo's brand to append '*'s to everything he posts).

The central puzzle of this paper is on the output side: it concerns whether, despite his minimal creative effort, Theo—and if not Theo, who or what?—should get artistic credit for *Hippie Philosophers**. But there's also a relevant question on the input side, which it will be useful to consider first. This is whether any of the artists who created the millions of images that Cube-E ingested (or the people or organizations that hold copyright to these images) have grounds for complaint. None of these artists granted consent, received compensation or were credited for their role in the production of *Hippie Philosophers**. But why should we worry about this regarding *Hippie Philosophers** when we clearly don't for *Hippie Philosophers*?

Hippie Philosophers,* made by Dall-E with a prompt from the author.[4]

To sharpen this question, let's assume that there are strong aesthetic similarities between *Hippie Philosophers* and *Hippie Philosophers**. In fact, for our purpose we can assume that they're formally and expressively identical—not only are they visual doppelgängers, but audiences interpret them as expressing the same things about the late 60's, hippie culture, the nature of art, and so on. For our purposes, we can also assume similarity between the database of images that Cube-E ingests and the large collection of images that influenced Arta's style.[5] So, given these assumptions, where's the difference? Even if we don't know how to assign credit, why should we withhold it for *Hippie Philosophers**, when we clearly don't for *Hippie Philosophers*.

A first possible difference is that while Arta copies and draws upon specific artistic influences in the process of developing her general artistic style, this style is more or less stable by the time she comes to paint *Hippie Philosophers*. And one might argue that the same cannot be said for Cube-E, which seems simply to copy the ingested

[4] There is currently no possibility of legal copyright to this image (see Moore & Frankel 2024). Moreover, if the central argument of this paper is sound, this image cannot be artistically credited to any person or thing.

[5] Arta's influences should properly include not only those images she absorbed directly through viewing and copying, but also those images that influenced the creation of these direct influences, influences on these secondary influences, and so on.

images, mechanically disassembling and reassembling them into combinations that allow it to respond to a prompt.

In reply to this, a distinction between the training and the generation of a specific image can also be drawn for deep learning systems like Dall-E, Stable Diffusion and our fictional, Cube-E. By feasting on millions of images (and tags that go with these images), Cube-E "learns" the general formal aspects of cubist imagery. It then puts its "knowledge" of the forms internal to this artistic sub-genre, along with visual hippie stereotypes, to work in generating *Hippie Philosophers**. Perhaps this distinction between general stylistic training and the generation of a specific image can't always be cleanly drawn for Cube-E. But neither can it be for Arta, who might sometimes be so moved by a new influence that she absorbs and integrates a novel (to her) style in the process of generating a new image of her own.

A second possible difference is that, during the learning period, Arta absorbs and filters artistic influences through a fully conscious, perceptive, reflective, unified, and dynamic human experience. Arta is, after all, an artistically sensitive human being. She takes in, and mindfully copies by hand, the images she encounters in a way that's filtered, as are all of Arta's perceptual experiences, through her ever-changing world view—that is, through a conceptual scheme that crucially includes her developing sense of artistic style and expression. Cube-E does no such thing. As I'll discuss further below, Cube-E doesn't bring to the training or generative process anything like conscious perceptual experiences, much less experiences that are shaped and informed by a complex, reflective and normatively responsive conceptual scheme.

True enough— Cube-E might be said to have "expertise" or "intelligence" only in an extended, anthropomorphized, and domain-specific way. Nevertheless, through deep learning, generative AIs *are* able to abstract formal visual patterns and connect them with the abstracted natural language patterns used to process prompts. There's no denying that, however one describes it, in producing *Hippie Philosophers**, Cube-E does something impressively synthetic and generative—something in the vicinity of creativity. In any case, this thought experiment supports the view that none of the artists who created the images that are input to a generative AI like Dall-E should be given artistic credit for specific outputs, such as *Hippie Philosophers**.

4 The no-artist argument and the antinomy of artificial art

So, if we accept that *Hippie Philosophers** (henceforth *HP**) is credit-worthy, to whom, or to what should we assign this credit? There's no clear and unambiguous candidate, I think. And this gets to our central puzzle. In this section, I'll first give *prima facie* motivation to an argument by cases to the conclusion that, in fact, *HP** and many AI-generated images lack an artist. Then I'll show how this "no-artist argument" generates my titular antinomy.

The no-artist argument proceeds by cases. *Prima facie* at least, none of four plausible candidates for credit—the input artists, the AI itself (Cube-E), its programmers, and the prompter (Theo)—count as the artist of *HP**. If this list of candidates is exhaustive then *HP** seems to be an artwork without an artist.

So, what about the artists of the images that Cube-E ingests? Should they be credited with creating *HP**? Well, if the Arta–Theo argument I gave in the last section is sound, then we should no more count the creators of the input images as having created *HP** than we should credit Picasso et al., instead of Arta, with having created the specific painting *Hippie Philosophers*.

What about the programmers of Cube-E? Generative AIs are, of course, amazing accomplishments in their own right, and the people that design and program them should be credited and well-compensated for creating them. But this doesn't mean that they should also be credited with the specific outputs their programs generate. If Cube-E were "explicitly" programmed—that is, if its programmers coded in rules that explicitly mentioned the visual details, norms of composition, and stylistic details about cubist painting—then we might credit the programmers on the grounds that the generation of the specific image, *HP**, was sufficiently guided by the programmers' knowledge of art. But image-generating AIs don't work this way. Through machine learning, the AIs educate themselves. The programmers don't need to know anything at all about cubism, about hippies, or even about visual art. So, holding that programmers should be credited with *HP** is no more plausible than holding that Henri Cartier-Bresson's famous photograph, *Place de l'Europe Gare Saint Lazare,* should be credited to the designers of his Leica camera, or of the camera in general.

What about crediting the AI itself—Cube-E in our example? One reason against doing so is that Cube-E isn't human—it's not a sentient or even living being. Under current US law, this would be reason enough to deny Cube-E legal standing for copyright to *HP**.[6] But even if we put aside any speciesist, sentientist or vitalist considerations, Cube-E doesn't have the type of full intentional awareness and self-monitoring that seems required for artistic creativity. Someday, perhaps someday very soon, AIs will pass a sufficiently robust Turing test to count, on some theories, as fully intentional and autonomous agents. But that day isn't here yet. The conceptual challenge of our current moment (2025) is that generative AIs are capable of stunning, but nevertheless domain-specific and still limited, accomplishments in specific areas—such as image generation. They are at best "partial agents", and our current conception of art and art-making, grounded as it is in the one artist-one artwork paradigm, requires that artists be intentionally sophisticated and normatively responsive agents. By this I mean that artists paradigmatically have a concept of art under which they engage in their art-making activities, and that these in turn are shaped and guided by sensitivity to external and internalized aesthetic norms and considerations. (I'll elaborate further on this aspect of our paradigm in Sect. 6.) For this reason, it seems implausible to count Cube-E as the artist of *HP**.

Finally, what about the prompter, Theo? It's true that Theo provides a half-baked idea when he types in his prompt, but he doesn't seem to do anything artistic. Unlike the programmers, Theo must have some minimal awareness of hippies and of cubism to even come up with the prompt. But he needn't know any of the formal stylistic details that go into cubism—nothing about lines, coloring patterns, representational deconstruction, and so on. And Theo needn't know much about hippies, about how

[6] Although this issue is still contested, a series of judgments in the famous monkey selfie dispute of the mid-2010's held that only works created by a human can be copyrighted, at least in US law.

they look, or about how many there should be in the image. All of the formal and representational details are taken care of by Cube-E. So, Theo seems not to do enough to be plausibly credited as the artist of *HP**.[7]

In the next few sections, I'll take up several ways one might try to push back against this argument, but what I hope I've shown in this section is that, *prima facie* at least, no obvious candidate can be credited as the artist of *HP**. And if this no-artist argument is sound, then it leads straightaway to our conceptual puzzle. *Hippie Philosophers** certainly seems like a work of art, even if it's not a great one. We might engage with it aesthetically in the same ways that we treat the formally indistinguishable *Hippie Philosophers*, which is indisputably a work of art. But artworks are created by artists like Arta; they are products of a creative process that centrally involves an intentional agent or agents. And if the conclusion of the no artist argument is sound, no such artist created *Hippie Philosophers**.

Using *Hippie Philosophers** as a representative example, we might articulate this deep tension in our conceptual scheme in the following three incompatible propositions:

A. *Hippie Philosophers** is a work of art.
B. Artworks are created by artists.
C. No artist created *Hippie Philosophers**.

This is the titular antinomy of artificial art. I call it an "antinomy" because I don't think any empirical or theoretical discovery concerning our current concepts and artistic practices will resolve it. If I'm right, the antinomy can only be side-stepped by revising the concepts that surround our practices of art. And any such revision will, I suspect, be a matter of choice—an implicit but institutionally supported decision that's driven by pragmatic and social considerations surrounding these practices. But before I elaborate on this speculation, I'll consider three different ways that one might try to resolve the antinomy.

5 Pushback #1: prompter as institutional artist

Perhaps we were too quick to dismiss Theo as the artist behind *Hippie Philosophers**. It's true that Theo doesn't do anything traditionally artistic: he does nothing materially, and he provides no artistic details to the generation of *HP**. Nevertheless, Theo is a fully intentional and autonomous adult human with a concept of visual art. So

[7] I'm not focused here on the way that artistic credit is implemented in copyright law. But for what it's worth, there's a legal distinction between an *idea* and the *expression* of that idea that supports this conclusion. Suppose you come up with a creative idea for a painting, and you then share this idea with me over coffee, including detailed thoughts about how the painting should be rendered artistically. If I rush back to my studio and execute your idea with paints on canvass, it is I, not you, who have copyright to the image. This is because I'm the one who "expressed it"—the person who brought the painting into existence materially. So, returning to our main example, not only does Theo not formulate or articulate any artistic details, but even if he did, Theo wouldn't get legal credit, since he doesn't materially express *HP**. So, if I'm right, Theo counts as the artist of *HP** neither conceptually nor legally.

perhaps in selecting and posting an image that he finds compelling, Theo effectively transforms *HP** into a work of art.

This line of thought might get support from Dickie's well-known institutional theory of art.[8] The institutional theory holds, very broadly, that something counts as a work of art largely because it gets established as such by social work an artist carries out in their "art world"—that is, in a set of social and norm-driven institutions, such as galleries, art critics, shared conceptions of artistic genre, and so on. Because this social work needn't be very material, the institutional theory gives an excellent account of ready-mades like Marcel Duchamp's seminal *Fountain*. According to the theory, Duchamp famously turned, or "transfigured", an ordinary urinal into the work of art, *Fountain*, not through any physical manipulation of the urinal, but rather by getting people in his artworld to consider and evaluate the urinal as a work of art. If the institutional theory can make good sense of why Duchamp's *Fountain* counts as an artwork, and Duchamp its artist, perhaps the account can be applied to Theo: in selecting and posting, Theo transforms *HP** into an artwork by operating within the implicit social norms and institutions of his on-line artworld.

I won't argue here that the institutional approach *can't* be fleshed out so as to support Theo as artist of *HP**, but I'll sketch several reasons for thinking that this would be a revisionary rocky road. First, even if we put aside controversy about institutional theories generally, and also accept that they make good sense of ready-mades like *Fountain*, there are, I think, significant disanalogies with *HP**. There are two ways in which we might take Duchamp to have transfigured a urinal into *Fountain*. First, he might have got us aesthetically to appreciate formal features of the object—its sheen and shape, for example—that we might not appreciate when we regard it functionally, that is as a urinal. Second, Duchamp might have institutionally embedded the urinal in such a way as to express a conceptual statement or puzzle about the nature of art itself: he got viewers to question what exactly they were encountering, and how they should evaluate it in an artistic setting such as a museum. It's this second respect of transformation that gives ready-mades their significance in twentieth century art: ready-mades were understood more as works of conceptual art than as odd sculptures.

The intent and effect of *HP** is clearly *not* to raise any conceptual puzzle about on-line art. So, the transformation that Theo effects, according to this institutional push-back, is more like the first way of understanding Duchamp's accomplishment. That is, Theo brings us to regard aesthetically and prominently the artistic features of something we wouldn't otherwise regard in this way. But what would this object be?

I have so far talked vaguely of transforming *HP** into a work of art, but what, more exactly, is transformed into what? We might claim that Theo transforms an arrangement of colored shapes and lines into a work of art, in the way that, to riff on an example from Putnam, I might declare that some ant trails we see in the sand can serve as a portrait of Winston Churchill.[9] But this would inaccurately capture Theo's

[8] See, for example, Dickie 1974, as well as Danto's seminal 1964.

[9] See Putnam 1981, Ch. 1. It's not clear, by the way, that I really can just declare the ant trails to be a portrait of Churchill. I might get away with this conversationally if there were a striking visual similarity between the trails and the shape of Churchill's face. This is the same sense in which I might ask if you

accomplishment. What Theo reacts to, selects and posts isn't merely an arrangement of colored shapes and lines, but is rather something that everyone, especially Theo himself, regards as having its representational and expressive features already built in. So, the thing Theo putatively transforms is most plausibly understood as an "interpreted image". To make sense of this possible transformation, we might, for the moment, carefully distinguish *HP**-the-image from *HP**-the-artwork. We can then understand the institutionalist as claiming that Theo transforms *HP**-the-image into *HP**-the-artwork.

However, if it's merely an image that's transformed, then this institutionalist response to the no-artist argument is too subtle by half. Perhaps Theo does accomplish something by getting people on-line to view the image more aesthetically—to appreciate and evaluate its artistic features—than they would if they thought the image was meant to serve as a brand icon, for example, or as an odd visual document of the summer of love. But this counts Theo as an "artist" in a very subtle and modest sense indeed.

Moreover, the view improperly situates the accomplishments of more involved prompters such as Charlie Engman.[10] Suppose Theo had put more thought, more detail, and repeated fine-tuning into the prompting that generated *HP**. In this case, we might be more inclined to credit him as artist, but it wouldn't be for institutionalist reasons—that is, it wouldn't be because Theo carried out a more significant transformation of image to artwork. It would be because he was more involved in the creation of the original image.

These considerations are connected to a deeper point. Even if we could somehow pursue the institutional route to establish Theo as, in some sense, the artist of *HP**, we would only defuse the force of the antinomy as I've articulated it—as an antinomy about artists and works of art. But the conceptual puzzle that animates my antinomy goes beyond art; it bears on the creation of images more generally. Who created *HP**-the-image? None of our candidates—Cube-E, its programmers, the creators of the input images, and Theo—seems up to this task either. Yet, unlike the ant-trails in the sand, there is indisputably an image here, and it's one that cries out for a creator.

I end this section by raising a more general problem for the idea that a human prompter might be defended as a way out of the no-artist argument. This is the simple, end-run point that human prompters seem in principle dispensable in the creation of AI art. As an experiment, I asked Chat-GPT to suggest a good prompt to put into Dall-E. Chat-GPT replied, "cosmic symphony of planets as musical instruments" which I then typed into Dall-E, generating image #2. The result, *Cosmic Symphony,* seems just as much a work of art as *HP**, yet here the human (i.e., me) didn't provide a specific prompt.

[image #2]

notice a dragon in a particular cloud formation. But what would seem to ground the claim that the ant trails really were part of an image of Churchill would be if they stood in the right type of causal relation to Churchill. In fact, this is the type of point Putnam wanted to make with his example. In contrast with the ant trails, *HP*-the-image* does plausibly stand in the right type of causal relation to hippies, and to Cubist stylings. *HP*-the-image* inherits its representational and artistic features from the images it ingests, and they in turn from the intentionality of the artists that produced them.

[10] See the *New Yorker* profile of Charlie Engman (Wiley 2023).

Cosmic Symphony, made by Dall-E with a prompt from the author.

True enough, I might be credited with the second-order idea of using Chat-GPT to generate a specific prompt for Dall-E. But even this minimal human involvement can be finessed. Consider a self-prompting "Art Machine", which, for all I know, already exists somewhere. Every night, at the stroke of midnight an AI automatically self-prompts and generates a novel artistic image displaying it on a living room screen for passers-by to appreciate throughout the following day. There would be no question of attributing human credit for the Art Machine's specific outputs, since no humans are involved at any stage.

6 Pushback #2: AI as creative agent

Maybe we were too quick to rule out the AI, Cube-E, as the artist behind *HP**. One reason for regarding *HP** as a work of art is that it strikes us as creative—as the novel and aesthetically valuable output of a process that is seemingly sensitive to the expressive dimensions and genre nuances of visual art. We've seen that the locus of this creativity isn't Theo. But neither does it lie in the programmers or the input artists. So, we might reconsider whether, despite its non-human status, the AI, Cube-E, is the creative force behind *HP**, and so to be credited as its artist.

This line of response turns on whether we really can count generative AIs like Dall-E, Stable Diffusion and Midjourney as artistically creative. In this section, I'll briefly sketch two reasons for optimism on this front, before finding pessimism in the thought that the current generation of AIs are not yet fully autonomous agents. The upshot is that, in order to succeed, this line of thought would involve revising our concept of creativity and its cognates.

The two reasons for optimism emerge from reflecting on the nature of human creativity, and finding striking similarities in the workings of generative AIs. First, we might look at Margaret Boden's influential taxonomy of creativity. Boden distinguishes three forms of creativity—combinatorial, exploratory and transformational.[11] Boden describes *combinatorial* creativity as "the unfamiliar juxtaposition of familiar ideas." In the case of visual art, this would be when artistic ingredients—visual elements and patterns—are combined in novel ways. And in fact, generative AIs seem to deconstruct input images into artistic elements, and then to re-combine and synthesize them in a computationally guided way that strongly parallels this variety of creativity.

A second species of creativity is *exploratory* wherein "some familiar style/space is explored by generative processes that construct novel structures informed, and limited by the constraints defining them." The human artist, Arta, for example, has a well-developed sense of the forms that go into cubist imagery. Even if these constraints are vague and merely implicit—and even if Arta can't articulate them in words, she has a sense of what would and wouldn't count as a well-formed cubist image, and also of what makes for better or worse images in this style. It is with these cubist sensitivities in place, that Arta paints *Hippie Philosophers*.

The AI, Cube-E, by contrast, can't be said to have a "sense" of anything, much less an articulated world view. Nevertheless, it's capable of producing images through a process that seems exploratory in just this way. It receives a prompt that is novel to it (e.g., "Hippie philosophers in cubist style") and generates an image of hippie philosophers that abides by formal constraints characteristic of Cubism. One might object that a difference here is that the cubist genre constraints within which Cube-E fashions its images aren't known to Cube-E. But, as we just suggested, they needn't be known to Arta either, at least not explicitly. Cube-E absorbs these constraints implicitly through a process of deep learning on all those ingested images. What counts here is know-how, not knowledge-that.

Boden articulates a final notion of *transformational* creativity "wherein the novel structure does not fit into any known style". This is the type of creativity that applies to great "out of the box" scientific breakthroughs, as well as to artworks that point to entirely new styles or genres in the way, in 1907, Picasso's *Les Desmoiselles D'Avignon* ushered Cubism into being. Generative AIs don't yet seem capable of this type of "outside the box" image generation; and maybe they never will be. But many highly valued and original artworks—probably the vast majority—aren't transformational either. So, Cube-E and other generative AIs seem to be creative in both Boden's combinatorial and exploratory ways, even if they don't yet create in transformational ways. Two out of three 'aint bad.

There's a second promising analogy—one which holds between the cognitive structure of human creativity and the architecture of generative AIs. According to leading cognitive theories, human creativity plausibly involves two phases, a generating phase and an evaluative phase.[12] When we engage in a moment of creativity, we first generate (sometimes in imagination) sketchy ideas—pathways or rough

[11] See Boden 2010, p. 227–8. See also Boden 2014 and 2018 for further discussion of this distinction.

[12] See Stokes and Paul 2021.

schemata for a fully worked-out idea. We then select among candidates, evaluating them according to some sort of task-specific metric. Finally, we might edit and re-prompt in the process of coming up with a final thought. This is often done quickly, and beneath the surface of conscious awareness.

Tantalizingly, the architecture of many generative AIs involves two phases very much like this. First, there's a generative adversarial network (a GAN) wherein candidate ideas are put in competition with one another in the process of generating a restricted range of possibilities. And then there's a discriminating "clip mechanism" that brings the prompt to bear in a process of evaluating and chucking out some of these possibilities, and then re-prompting. The parallel with human creativity is striking.

So, reflecting both on Boden's taxonomy of creativity, as well as upon the structural "two-phase" process analogy, we might be more willing to situate the locus of artistic creativity in Cube-E. However, when we remember that generative AIs don't yet approach the status of fully autonomous agents, we might more cautiously understand these attributions of creativity as charitable projections on our part.

The central challenge in attributing artistic creativity to an AI derives, I think, from the "one artwork–one artist" paradigm I articulated at the outset: starting with material ingredients, stylistic influences, and some sort of expressive goal, a lone human artist launches an autonomous and intentionally adjusted creative process that ends in a unified artistic expression—an artwork. Two connected features of this paradigm challenge the attribution of creativity to the current generation of generative art AIs: the capacity for robust intentionality and the capacity for normative responsiveness. I'll briefly explain what I mean by each.

By robust intentionality, I mean the ability to represent to oneself, and to appropriately adjust not only to the artistic details and dimensions of an art genre, but also to the expressive goals and purpose of the artistic undertaking itself. Ideally, the artist is explicitly reflective about the nature of their artistic practice, and even about the nature of art. Of course, many human artworks are created non-reflectively or even entirely unconsciously, with only implicit representation of process and artistic detail. A famous example is the first thirteen lines of *Kubla Khan*, which came to Coleridge in an opium-induced dream-state (at least as that episode is described through the 19th-century lens of romantic genius). Less spectacular cases happen all the time when an artist "gets in the flow" or distractedly "tosses off" a work. Indeed, in performance art, it's often best not to be fully reflective in the moment, since performance and improvisation suffer from excessive self-scrutiny. And of course, many great artists never explicitly reflect on the nature of art itself. But in all these cases, the artist clearly has the *capacity* to be fully reflective about their artistic undertaking, and usually has been reflective in the past, when learning, fine-tuning, and practicing their craft. The current generation of AIs clearly do not (yet) have this capacity for robust intentionality, and so, the works they produce don't clearly and unproblematically satisfy our received concept of creativity.

A second, connected feature of the one-artwork–one-artist paradigm is the capacity for normative responsiveness. By this I mean the capacity to respond and adjust to praise, blame, and other normative—in this case, aesthetic—constraints and attributions. There is a use of 'create' that is merely causal, as when we say that the

moon's gravity creates ocean tides. But the way we use 'create' in the context of art-making—that is, 'create' as it figures in claim B of the antinomy—seems minimally to require that the artwork be brought about through a process initiated by a *creator* who is more than a mere causal source. A minimal condition on being an artistic creator is that the agent has an intention to succeed in making something artistic. And this intention has a normative dimension. The intention needn't lead to a good work of art, but the creator must have a sense of what would count as success and failure in their artistic undertaking. More than this, the artistic creator is normally moved to orient their training, planning, and activity toward succeeding in the creative process. To be fully responsive to the demands of creation in the domain of art requires, I suggest, the capacity to be normatively responsive more generally.

True, there may be occasions in which an agent might not genuinely intend to succeed at creating art, as in a student-assignment started under duress. But even here, the agent must be normatively correctable, even if these demands are rejected in the moment. True as well, we sometimes compartmentalize our attributions of creativity. We might say that someone is a highly creative musician, for example, even if they are unimaginative, rigid, and conformist in all other areas of life. But here too, I think, we require that the agent be normatively responsive in general. A domain-limited attribution of creativity is, I'm suggesting, made against an aretaic ideal of general creative excellence.

My claim, then, is that artistic creativity is a prototype concept—one grounded in the one-artist–one-artwork paradigm. And the prototypical artist—the creator to whom we want to give credit—is at least capable of what I have called robust intentionality and normative responsiveness. The paradigm shapes our expectations concerning the intentions, accountability, and aesthetic responsibility of an artistic creator. As with any prototype concept, we might apply the label "artist" to cases that don't fully satisfy all prototypical features built into the underlying concept. But when we do, we tend to feel a degree of unclarity or unease about this attribution. If I'm right about this, and also about the fact that, sophisticated as they are, current generative AIs are neither robustly intentional nor normatively responsive, then we can't clearly, comfortably, or unproblematically credit them with creating a work like *HP**.

Of course, we might change the way we use "artistic creativity" so that it no longer invokes this concept. Perhaps we will drop or relax the prototypical features currently built into our concept. But this wouldn't be to comfortably count an AI like Cube-E as creative under our current conceptual scheme.

7 Pushback #3: artistic collectives

A third line of pushback against the no-artist argument is to hold that, although none of the individual candidates counts as the creator of *HP**, the work is nevertheless the product of a collective artistic undertaking. Maybe we can say that Theo and Cube-E collectively created *HP**, and hold this in a way that credits Cube-E as more than

merely an artistic tool, such as a camera.[13] Perhaps "Theo-cum-Cube-E" is an artistic collective—a group entity that is artistically responsible for *HP**? If so, we might reject the no-artist argument (and proposition C of the antinomy) on the grounds that our list of candidates has not been exhaustive: we haven't considered the possibility that Theo and Cube-E are to be fully credited as a group, and partially as individuals.

Attributing artistic creation to a collective is something we do all the time. The Beatles—John-cum-Paul-cum-George-cum-Ringo—are collectively credited with the album *Sgt. Pepper's Lonely Hearts Club Band*, even though their exact contributions were distinct, and often credited as such. Likewise, hundreds of individuals are credited in the artistic production of large-scale works like the film *Dune*. But in these cases, we tend to assume that each credited contributor played a socially defined creative role that bears on the work's *artistic* features. Moreover, each contributor played a part in shaping the expressive work in a way that requires the capacity for robust intentionality and normative responsiveness—capacities that I argued in the last section are absent from current generative AIs.

Take the Beatles. While we might not credit Ringo with writing any of the songs on *Sgt. Pepper*, we still count him among its artistic creators. He contributed artistically by refining drum parts in ways that support the album's overall expressive goals, and participating in group discussion about the music. By contrast, we do not consider the executives and engineers who built EMI Studios, or even their long-time manager, Brian Epstein, to be members of this artistic collective despite the fact that their causal contributions were necessary for the album to exist in the form it does. Without their involvement, the album—*that* album—might never have been made, or might have sounded very different. Still, their roles weren't artistic in nature; these people didn't contribute in any intentional and responsive way to shaping the expressive content of the album.

This view helps us address edge cases. Should George Martin—the so-called "fifth Beatle"—count as part of the artistic collective behind *Sgt. Pepper*? While not a member of the band in the commercial or contractual sense, he played a major artistic role: producing the album, engineering its sound, mixing the tracks, and even performing on occasion. Given the degree of creative input he exercised, there's a strong case for including him in this artistic collective, even if he doesn't count as one of the Beatles generally (he wasn't as marketable, and he didn't play on-stage, for example).

Other cases are even more delicate, and push up against vagueness. There seems to be consensus now that Rembrandt was the actual painter of the famous painting, *Polish Rider*, in the Frick collection. But suppose, as some have argued, that his apprentice, Willem Drost, actually painted the much-admired face. Should Drost be considered part of an artistic collective—Rembrandt-cum-Drost—that created *Polish Rider*? I think not, if Drost was merely painting the face in a way that Rembrandt sketched or outlined (although Drost might be credited with skillfully applying the paint). This is what we would say about William Suhr, one of the Frick's conserva-

[13] For a careful and nuanced attempt to flesh out this idea, see Khosrowi et al., 2023. As I understand it, the authors are there making a proposal about how we might *revise* our practice of attributing credit. So, as far as I can see, their prescriptive proposal is compatible with all that I say below.

tors, who apparently repainted some of the horse's hooves in the 1950s. On the other hand, if instead Rembrandt said only "paint a face on my polish rider" and let Drost at it, then Drost probably should be included in the artistic collective that created *Polish Rider*.[14]

What do we say of all the people who work on a film like *Dune*? I've just argued that our received practice is to attribute partial artistic credit to anyone who intentionally and responsively contributes to the artistic vision of either the film as a whole *or* to some relatively well-defined artistic task within the film-production. In the first category we would presumably include the director and screenwriter. In the second, we might include the lighting coordinator, if they were told, for example, to come up with lighting that brings out the alien desert atmosphere that the director wants to evoke. But there is vagueness here.[15] If the lighting coordinator was told simply "put a light here," then I wouldn't include them in the collective that properly gets artistic credit for the movie anymore than the caterer who is justly credited (and listed in the ending film crawl) for creating delicious lunches, but not for contributing to the artistic expression of *Dune*.

So where does all this leave the AI, Cube-E, in our central example? Should it be included in an artistic collective that lies behind *HP**? If I'm right about the one-artwork–one-artist nature of the prototype concept embedded within our artistic practices, then Cube-E doesn't count.[16] This is precisely where the AI case diverges from paradigmatic artistic collectives. It's not the non-human status of Cube-E that disqualifies it from artistic credit, but rather its absence of robust intentionality, responsiveness to norms, and ability to grasp and adjust to expressive goals. Cube-E does significant work in the production of the output image, executing an important subtask in mind-blowing ways. But even if Cube-E was causally central to the production of *HP**—*HP** wouldn't exist without it—that alone doesn't justify including it in the artistic collective. Cube-E contributes neither to the conception of the whole work, nor to any well-defined part of it in ways required by our current practices of credit attribution. Cube-E differs from the artistic operatives of *Dune*: even if the film's costume and lighting designers just carried out artistic sub-tasks, they did so in a robustly intentional and normatively responsive manner. Their full agency is a requirement for crediting them with partial, role-specific credit. Even a sophisticated, automatic camera lacks full agency, and for this reason we refrain from giving it even partial artistic credit for any photograph we might use it to take. Art-aiding AIs are much more capable, and much more generative than cameras and the other artistic tools we're familiar with. But as of now, they don't have the robust intentionality and normative responsiveness required in order to be given any artistic credit under our current conceptual scheme.

[14] See Glueck 1997 for the historical details I'm drawing on.

[15] There is also a vague boundary between explicitly working on a work's "whole vision" and carrying out just a well-defined and delegated artistic part of this. A cinematographer might lie on this boundary. Fortunately, we needn't sharpen this distinction for my purposes here.

[16] And in our central example, Theo wouldn't count either. So, if I'm right, the no-artist argument stands: there's *no* collective, not even the singleton collective consisting of lonely Theo that plausibly counts as the artist of HP*.

8 Conclusion

Everything I've argued here applies to our *received* practices of granting artistic credit, and to the concepts they ground. My project has been one of descriptive analysis, not prescriptive, conceptual engineering. Our practices and concepts, along with their vague boundaries, have evolved among us humans—cognitively, socially, and physically wired as we are. They didn't evolve to accommodate the expressive behavior of non-humans, like monkeys, or of alien forms of intelligent expression. It's no wonder then that they can't smoothly handle novel, art-effecting technologies like generative AIs.

As new technologies and practices develop, we humans need adjust our ways of understanding and evaluating art. And we reshape the cultural, commercial, and legal institutions that surround new art-making. Along with the camera came the emergence of photography, but also changes in the practice of traditional portraiture. The invention of synthesizers and sound recording technology transformed not just musical composition and performance, but also the ways we distribute and experience music. Video technology has given birth to video art and changed filmmaking. And so on. So, as AI super-tools are increasingly integrated into our expressive practices, and as they create pressing practical problems, such as within intellectual property law, our conceptual-scheme and our practices of art-making themselves will need to shift.

The puzzle I've developed in this paper reveals a core conceptual conflict that arises as we try to apply to AI-art our *received* way of talking, thinking, and normatively engaging with art. We might—indeed, we probably will—choose to *revise* these ways, either through a gradual and implicit shift within our artistic cultures or with the aid of explicit changes in surrounding institutions, such as museums, galleries, prizes, art-markets and copyright regimes. Exactly which revisions will best suit our changing artistic practices, and the larger society in which they're embedded, is an important further question. Answering this question will be a matter not of discovery, but of implicit and explicit social choice.

Acknowledgments This paper draws on some of the central ideas in Moore & Frankel 2024. I thank my co-author, Simon Frankel, for allowing me to repeat and develop them here. I also thank Jan Michel for his unflagging encouragement, patience, and expert philosophical and editorial advice. Finally, I thank audiences at the collegial Artificial Researchers and Scientific Discoveries Conference in Fall 2023, as well as an audience at the AI in the Liberal Arts series at Amherst College in Spring 2024.

References

Boden, M. (2010). *Creativity and Art: Three Roads to Surprise*. Oxford University Press.

Boden, M. (2014). Creativity and Artificial Intelligence. In E. Paul & S. Kaufman (Eds.), *The Philosophy of Creativity* (pp. 224–244). Oxford University Press.

Boden, M. A. (2018). *Artificial Intelligence: A Very Short Introduction*. Oxford University Press.

Danto, A. C. (1964). The Artworld. *Journal of Philosophy, 61*(19), 571–584.

Dickie, G. (1974). *Art and the Aesthetic: An Institutional Analysis*. Cornell University Press.

Grace, G. (1997). Inside Art. *New York Times*, October 24, 1997, https://www.nytimes.com/1997/10/24/arts/inside-art.html

Moore, J., & Frankel, S. (2024). Creativity, Credit and Copyright in the Age of Artificial Art. *Journal of Aesthetics and Art Criticism*, *82*(3), 265–277.
Khosrowi, D., Finola F., and Elinor C. (2023). Diffusing the Creator: Attributing Credit for Generative AI Outputs. *Proceedings of the 2023 AAAI/ACM Conference on AI, Ethics, and Society* 8: 890–900.
Putnam, H. (1981). *Reason, Truth, and History*. Cambridge University Press.
Stokes, D. & Elliot P. (2021). Computer Creativity Is a Matter of Agency. *IAI News*, November 11, 2021. Institute of Arts and Ideas.
Wiley, C. (2023). A Photographer Embraces the Alien Logic of A.I. *The New Yorker*, May 8, 2023.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.